# Grade 10 – Term 1
# Decision Trees with Staged Filtering: C9 Practice

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## Evaluated criteria

**C9** Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

## 1 Introduction

The summary tables provide four statistics for each alternative, all in thousand USD and rounded to two decimal places. Use these supplied statistics directly; you are not required to recalculate them from the payoff matrices.

- **Maximax:** select the highest Max Payoff (the alternative's maximum payoff).
- **Maximin:** select the highest Min Payoff (the alternative's minimum payoff).
- **Minimax Regret:** select the lowest Max Regret. Regret in each state is the best payoff among all six original alternatives in that state minus the alternative's payoff; Max Regret is the largest of these regrets.
- **EMV (Expected Monetary Value):** select the highest EMV, calculated using the stated probabilities as payoff weights.

Apply each problem's filtering rules sequentially. Compare only alternatives retained from the preceding stage; eliminated alternatives cannot reenter. All original summary statistics remain fixed: do not recalculate any statistic after eliminating alternatives. When asked to keep the best $k$ alternatives, retain every alternative tied at the cutoff, even if more than $k$ remain. At the final stage, report a tie explicitly if multiple alternatives share the best value.

Represent each process in a four-column filtering table: # Step, Description, Left Alternatives, and Eliminated Alternatives. Step 0 lists all six alternatives. For each subsequent stage, record the criterion, filtering rule, relevant numerical values, and alternatives retained or eliminated at that stage. Identify the final selected alternative (or tied alternatives) in the last row. The problems use two to four filtering stages and vary the criteria applied.



# 2 Decision Trees with Staged Filtering

**Problems:**


2.1. City logistics policy

2.2. Regional distributor expansion

2.3. Trade finance guarantees

2.4. Insurance portfolio mixes


## 2.1 City Logistics Policy

### Problem Description

A city logistics office must choose one policy among $B_1, B_2, B_3, B_4, B_5, B_6$. Payoffs (thousand USD) and probabilities are:

| State | Probability | $B_1$ | $B_2$ | $B_3$ | $B_4$ | $B_5$ | $B_6$ |
|---|---|---|---|---|---|---|---|
| $T_1$ | 0.50 | 95 | 90 | 110 | 85 | 100 | 92 |
| $T_2$ | 0.30 | 70 | 75 | 60 | 80 | 65 | 78 |
| $T_3$ | 0.20 | 30 | 40 | 10 | 55 | 35 | 20 |

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs. All summary values are in thousand USD, to two decimal places.

| Alternative | EMV | Max Payoff | Min Payoff | Max Regret |
|---|---|---|---|---|
| $B_1$ | 74.50 | 95.00 | 30.00 | 25.00 |
| $B_2$ | 75.50 | 90.00 | 40.00 | 20.00 |
| $B_3$ | 75.00 | 110.00 | 10.00 | 45.00 |
| $B_4$ | 77.50 | 85.00 | 55.00 | 25.00 |
| $B_5$ | 76.50 | 100.00 | 35.00 | 20.00 |
| $B_6$ | 73.40 | 92.00 | 20.00 | 35.00 |

Filtering rules (all values in thousand USD; retain ties at every cutoff): (1) EMV: keep the 4 highest EMV values. (2) from those, Maximin: keep the 2 highest Min Payoff values. (3) from those, Minimax Regret: select the lowest Max Regret value.

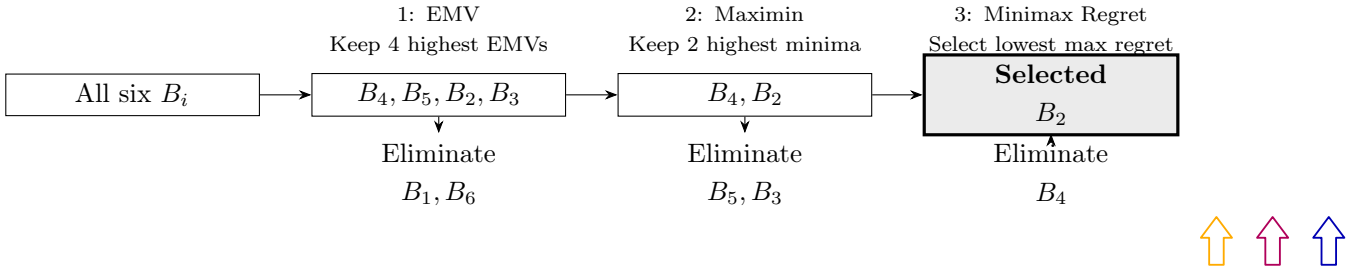
### Questions/Tasks



**C9** Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

**Solution**

| # Step | Description | Left Alternatives | Eliminated Alternatives |
|---|---|---|---|
| 0 | Initial alternatives | $\{B_1, B_2, B_3, B_4, B_5, B_6\}$ | — |
| 1 | EMV: keep the 4 highest EMV values. | $B_4(77.50)$, $B_5(76.50)$, $B_2(75.50)$, $B_3(75.00)$ | $B_1(74.50)$, $B_6(73.40)$ |
| 2 | Maximin: keep the 2 highest Min Payoff values. | $B_4(55.00)$, $B_2(40.00)$ | $B_5(35.00)$, $B_3(10.00)$ |
| 3 | Minimax Regret: select the lowest Max Regret value. | **Selected:** $B_2(20.00)$ | $B_4(25.00)$ |




## 2.2 Regional Distributor Expansion

### Problem Description

A regional distributor must choose one expansion plan among $X_1, X_2, X_3, X_4, X_5, X_6$. Payoffs (thousand USD) and probabilities are:

| State | Probability | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ |
|---|---|---|---|---|---|---|---|
| $U_1$ | 0.30 | 130 | 120 | 140 | 135 | 125 | 118 |
| $U_2$ | 0.25 | 90 | 95 | 80 | 100 | 88 | 92 |
| $U_3$ | 0.25 | 60 | 70 | 55 | 85 | 65 | 75 |
| $U_4$ | 0.20 | 20 | 40 | 10 | 60 | 45 | 35 |

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs. All summary values are in thousand USD, to two decimal places.

| Alternative | EMV | Max Payoff | Min Payoff | Max Regret |
|---|---|---|---|---|
| $X_1$ | 80.50 | 130.00 | 20.00 | 40.00 |
| $X_2$ | 85.25 | 120.00 | 40.00 | 20.00 |
| $X_3$ | 77.75 | 140.00 | 10.00 | 50.00 |
| $X_4$ | 98.75 | 135.00 | 60.00 | 5.00 |
| $X_5$ | 84.75 | 125.00 | 45.00 | 20.00 |
| $X_6$ | 84.15 | 118.00 | 35.00 | 25.00 |

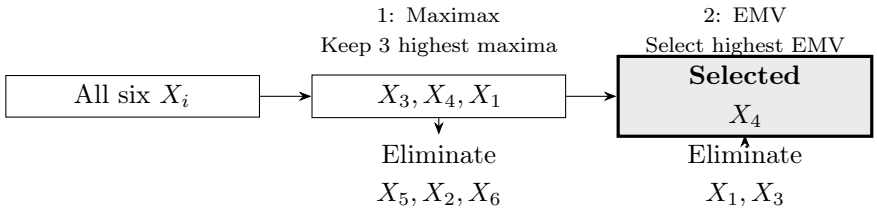
Filtering rules (all values in thousand USD; retain ties at every cutoff): (1) Maximax: keep the 3 highest Max Payoff values. (2) from those, EMV: select the highest EMV value.

**Questions/Tasks**

**C9** Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

**Solution**

| # Step | Description | Left Alternatives | Eliminated Alternatives |
|---|---|---|---|
| 0 | Initial alternatives | $\{X_1, X_2, X_3, X_4, X_5, X_6\}$ | — |
| 1 | Maximax: keep the 3 highest Max Payoff values. | $X_3(140.00)$, $X_4(135.00)$, $X_1(130.00)$ | $X_5(125.00)$, $X_2(120.00)$, $X_6(118.00)$ |
| 2 | EMV: select the highest EMV value. | **Selected:** $X_4(98.75)$ | $X_1(80.50)$, $X_3(77.75)$ |




## 2.3 Trade Finance Guarantees

**Problem Description**

A trade finance agency chooses one guarantee policy among $G_1, G_2, G_3, G_4, G_5, G_6$. States and probabilities are shown below. Payoffs are in thousand USD. Policies $G_4$ and $G_5$ balance upside and downside because they diversify by sector, while $G_3$ is highly cyclical.

| State | Probability | $G_1$ | $G_2$ | $G_3$ | $G_4$ | $G_5$ | $G_6$ |
|---|---|---|---|---|---|---|---|
| $M_1$ | 0.45 | 100 | 94 | 114 | 90 | 104 | 96 |
| $M_2$ | 0.35 | 72 | 78 | 82 | 84 | 70 | 80 |
| $M_3$ | 0.20 | 32 | 44 | 46 | 56 | 38 | 26 |

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs. All summary values are in thousand USD, to two decimal places.

| Alternative | EMV | Max Payoff | Min Payoff | Max Regret |
|---|---|---|---|---|
| $G_1$ | 76.60 | 100.00 | 32.00 | 24.00 |
| $G_2$ | 78.40 | 94.00 | 44.00 | 20.00 |
| $G_3$ | 89.20 | 114.00 | 46.00 | 10.00 |
| $G_4$ | 81.10 | 90.00 | 56.00 | 24.00 |
| $G_5$ | 78.90 | 104.00 | 38.00 | 18.00 |
| $G_6$ | 76.40 | 96.00 | 26.00 | 30.00 |

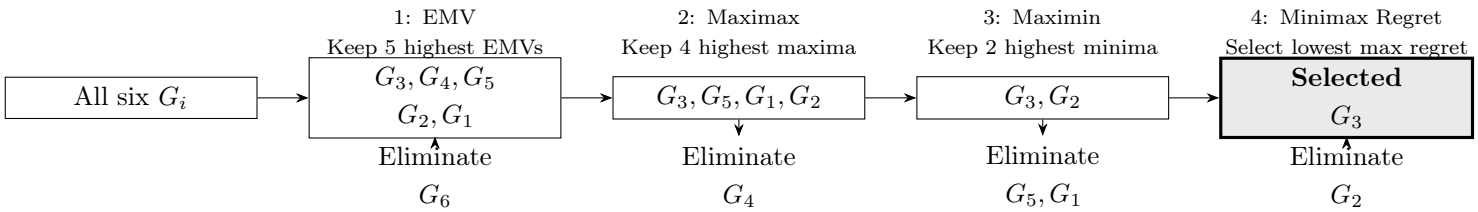
Filtering rules (all values in thousand USD; retain ties at every cutoff): (1) EMV: keep the 5 highest EMV values. (2) from those, Maximax: keep the 4 highest Max Payoff values. (3) from those, Maximin: keep the 2 highest Min Payoff values. (4) from those, Minimax Regret: select the lowest Max Regret value.

### Questions/Tasks

**C9** Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

### Solution

| # Step | Description | Left Alternatives | Eliminated Alternatives |
|---|---|---|---|
| 0 | Initial alternatives | $\{G_1, G_2, G_3, G_4, G_5, G_6\}$ | — |
| 1 | EMV: keep the 5 highest EMV values. | $G_3(89.20)$, $G_4(81.10)$, $G_5(78.90)$, $G_2(78.40)$, $G_1(76.60)$ | $G_6(76.40)$ |
| 2 | Maximax: keep the 4 highest Max Payoff values. | $G_3(114.00)$, $G_5(104.00)$, $G_1(100.00)$, $G_2(94.00)$ | $G_4(90.00)$ |
| 3 | Maximin: keep the 2 highest Min Payoff values. | $G_3(46.00)$, $G_2(44.00)$ | $G_5(38.00)$, $G_1(32.00)$ |
| 4 | Minimax Regret: select the lowest Max Regret value. | **Selected:** $G_3(10.00)$ | $G_2(20.00)$ |




## 2.4 Insurance Portfolio Mixes

### Problem Description

An insurance broker considers six portfolio mixes $Y_1$ to $Y_6$ under four climate-risk states with probabilities $W_1$:0.30, $W_2$:0.25, $W_3$:0.25, $W_4$:0.20. Payoffs are in thousand USD. $Y_4$ is conservative and protects downside in severe-weather years, while $Y_3$ has the highest upside but a weak stress outcome.

| State | Probability | $Y_1$ | $Y_2$ | $Y_3$ | $Y_4$ | $Y_5$ | $Y_6$ |
|---|---|---|---|---|---|---|---|
| $W_1$ | 0.30 | 136 | 126 | 146 | 116 | 130 | 121 |
| $W_2$ | 0.25 | 94 | 98 | 84 | 103 | 90 | 95 |
| $W_3$ | 0.25 | 66 | 74 | 58 | 88 | 69 | 77 |
| $W_4$ | 0.20 | 28 | 46 | 14 | 64 | 48 | 38 |

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs. All summary values are in thousand USD, to two decimal places.

| Alternative | EMV | Max Payoff | Min Payoff | Max Regret |
|---|---|---|---|---|
| $Y_1$ | 86.40 | 136.00 | 28.00 | 36.00 |
| $Y_2$ | 90.00 | 126.00 | 46.00 | 20.00 |
| $Y_3$ | 82.10 | 146.00 | 14.00 | 50.00 |
| $Y_4$ | 95.35 | 116.00 | 64.00 | 30.00 |
| $Y_5$ | 88.35 | 130.00 | 48.00 | 19.00 |
| $Y_6$ | 86.90 | 121.00 | 38.00 | 26.00 |

Filtering rules (all values in thousand USD; retain ties at every cutoff): (1) EMV: keep the 3 highest EMV values. (2) from those, Minimax Regret: keep the 2 lowest Max Regret values. (3) from those, Maximin: select the highest Min Payoff value.

### Questions/Tasks

**C9** Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

### Solution

| # Step | Description | Left Alternatives | Eliminated Alternatives |
|---|---|---|---|
| 0 | Initial alternatives | $\{Y_1, Y_2, Y_3, Y_4, Y_5, Y_6\}$ | — |
| 1 | EMV: keep the 3 highest EMV values. | $Y_4(95.35)$, $Y_2(90.00)$, $Y_5(88.35)$ | $Y_6(86.90)$, $Y_1(86.40)$, $Y_3(82.10)$ |
| 2 | Minimax Regret: keep the 2 lowest Max Regret values. | $Y_5(19.00)$, $Y_2(20.00)$ | $Y_4(30.00)$ |
| 3 | Maximin: select the highest Min Payoff value. | **Selected:** $Y_5(48.00)$ | $Y_2(46.00)$ |

